UNIVERSITY OF TEHRAN

Faculty of Electrical & Computer Engineering · M.Sc. Control Engineering

Fault-Tolerant Attitude Control of Nanosatellites Using Reinforcement Learning

From spacecraft attitude-control fundamentals to the research gap

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Image: NASA/JPL-Caltech, NISAR in Earth orbit (artist's concept), image PIA26285.

ROADMAP

From the engineering problem to the research question

01

The problem

What a spacecraft must control, and why it fails

02

The proposal

A learned controller that is allowed to fail safely

03

Related research

Three recent directions of the state of the art

04

Gaps & new ideas

What is still missing, and what this thesis adds

05

Conclusion

Why the direction is worth pursuing

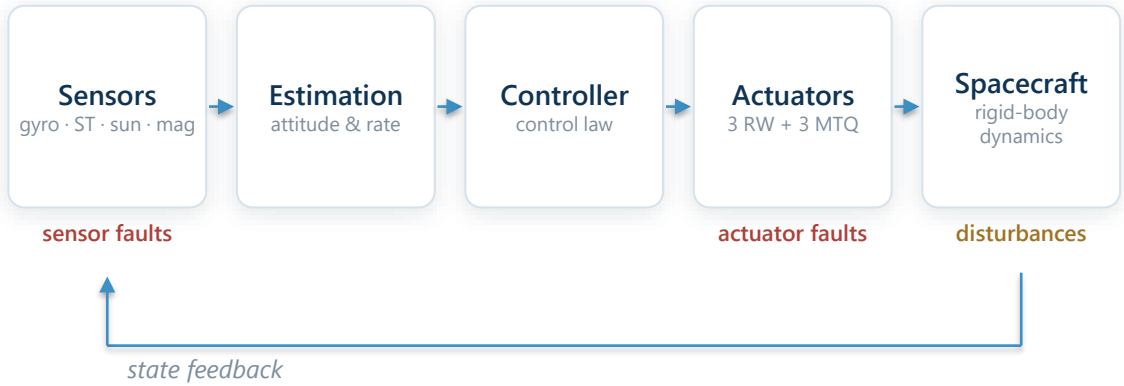
Ten minutes, five questions — one argument.

Attitude control: a loop that never stops

Point
hold the payload axis on its target — nadir, a star, a ground station.

Rotate
slew between targets inside torque and rate limits.

Reject
gravity gradient, drag, solar pressure, residual magnetic dipole.



ADCS estimates the attitude and acts on it — continuously, onboard, with limited power and computing.

Faults enter the loop in two places: **the measurements** and **the torque**. A fault-tolerant ADCS has to survive both.

Sources: NASA, State-of-the-Art of Small Spacecraft Technology — Guidance, Navigation and Control (2026); NASA SSRI Knowledge Base, Attitude Determination and Control.

Why fault tolerance is the hard part

Nanosatellites cannot be repaired, and they fly on commercial off-the-shelf parts.

1–12 kg — Mass and volume budgets shrink actuators, computing and redundancy.

COTS — Commercial parts cut cost, but widen uncertainty and failure rates.

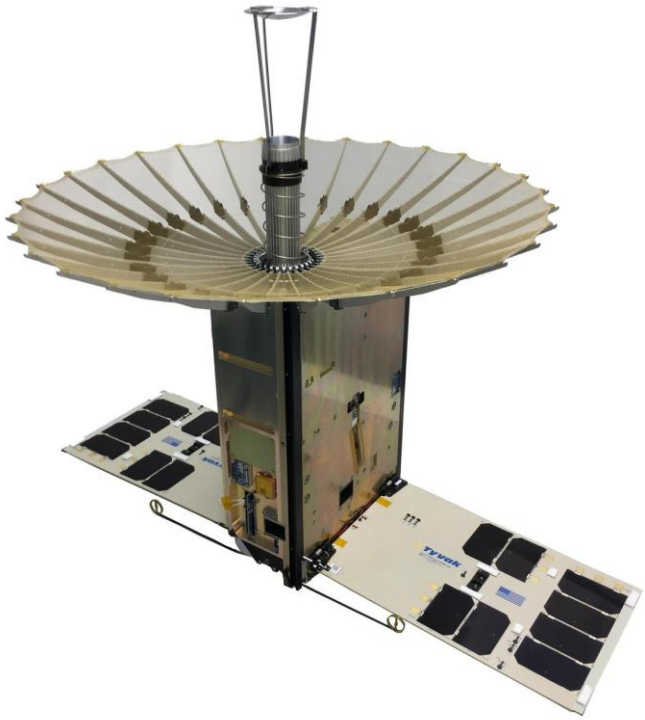
No servicing — A fault in orbit must be survived by software, not by a repair crew.

FAULT MODEL

$$\tau_{applied} = h \cdot \tau_{cmd}, \qquad h \in [0, 1]$$

continuous loss of authority — 20 %, 50 % or total — not a binary switch

Wheel friction · torque loss · encoder fault · wheel lock · sensor bias · dropout



6U CubeSat (RainCube) — the class this thesis targets.

Image: NASA/JPL-Caltech, RainCube 6U CubeSat, image PIA22457.

Can reinforcement learning deliver **adaptive, fault-tolerant** attitude control for a nanosatellite — and still keep pointing accurate when the system **is not the one it trained on**?

Unseen faults | **Safety guarantees** | **Onboard feasibility** | **Verifiable behaviour**

A controller is only useful if it survives the fault it did not expect.

A learned controller that is allowed to fail safely

Objective. Drive a nanosatellite from a random attitude and rate to nadir pointing — fast and precisely — and keep that performance when a reaction wheel degrades, locks, or fails in a way the policy never saw.

1 **Learn**

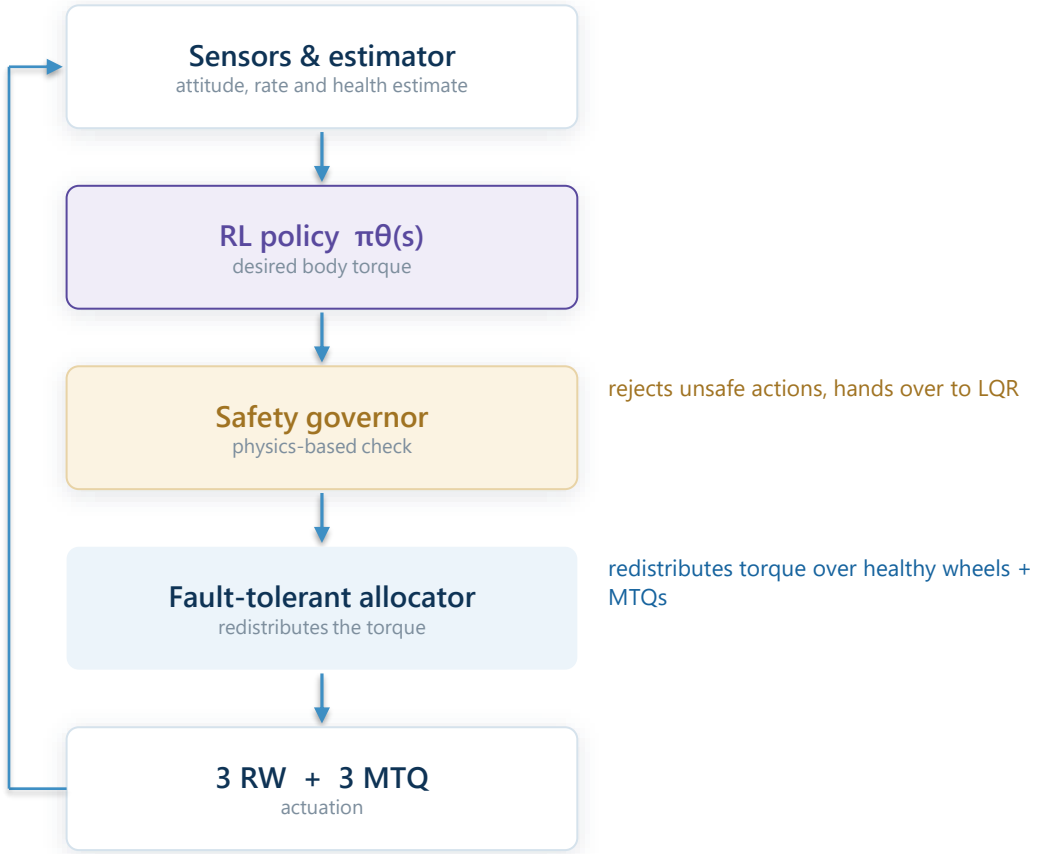
PPO / SAC on a nonlinear plant with random inertia, disturbances and noise.

2 **Never assume health**

The policy sees the estimate $\hat{h}$ — never the true health h.

3 **Fail safely**

A physics-based governor plus an LQR fallback — never a second network.



Validated in four stages — and tested on faults it never saw



Scenario	In training	Held out
Healthy spacecraft	trained	tested
Wheel at 50 % torque	trained	tested
Wheel dead / locked	—	key test
Unseen severity & combinations	—	key test

WHAT IS MEASURED

- settling time to nadir lock
- pointing error and success rate
- control effort and wheel saturation
- inference latency and memory on the target MCU

The claim is not "RL beats LQR". *The claim is graceful degradation where classical control has none.*

Three papers, three complementary answers

PAPER 1

Yang, Jin & Rao

Journal of the Franklin Institute, 2025

RL adapts online when the spacecraft model is unknown and actuators fail.

Adaptation

PAPER 2

Lu et al.

Aerospace Science and Technology, 2026

Safe RL keeps agile tracking inside operational constraints and actuator faults.

Safety

PAPER 3

Nagano & Schaub

AAS 25-637, 2025

Fault-aware training keeps mission scheduling resilient after wheel failures.

Resilience

Common question: *how can a learned controller remain useful when the real spacecraft does not behave as modelled?*

Papers: Yang, Jin & Rao, J. Franklin Inst. 362 (2025); Lu et al., Aerosp. Sci. Technol. 168 (2026); Nagano & Schaub, AAS 25-637 (2025).

Reinforcement Learning Based Attitude Fault-Tolerant Control of Spacecraft with Unknown System Model

PROBLEM

Classical fault-tolerant control needs an accurate model and known actuator behaviour — which real nanosatellites do not provide.

METHOD

- Offline: RL with optimal control gives a nominal policy.
- Online: a neural observer estimates the unknown dynamics and input gain.
- Recursive least squares updates the critic; a feed-forward term compensates additive faults.

The policy adapts online to actuator faults without a known system model.

WHAT IT LEAVES OPEN

- fault severities never seen in training
- sensor faults, not only actuator faults
- safety of the learned action itself
- cost of running the adaptation onboard

RELEVANCE TO THIS THESIS

Closest to this thesis in problem statement — and furthest from its deployment claim.

Yang, S., Jin, L. & Rao, J. (2025). Journal of the Franklin Institute, 362, 107741. doi:10.1016/j.jfranklin.2025.107741

Safe Reinforcement Learning Based Agile Satellite Attitude Control for Non-Cooperative Target Tracking with Operation Constraints and Actuator Faults

PROBLEM

Agile tracking must respect forbidden pointing zones, rate and torque limits — while an actuator is losing effectiveness.

METHOD

- Fault identification estimates effectiveness loss and additive fault.
- The task is written as a constrained Markov decision process.
- A worst-case soft actor-critic solves it, with faults inside the constraint set.

Fault tolerance, operational constraints and safety in one learned controller.

WHAT IT LEAVES OPEN

- computational weight for an embedded target
- verification of the safety claim
- faults outside the training distribution
- migration from workstation to flight hardware

RELEVANCE TO THIS THESIS

Shows that performance without constraints is not a result — constraints belong in the design.

Lu, W. et al. (2026). Aerospace Science and Technology, 168, 111238. doi:10.1016/j.ast.2025.111238

Enhancing Fault Resilience in RL-Based Satellite Autonomous Task Scheduling

PROBLEM

At mission level, wheel faults cut agility: reduced manoeuvres, safe-mode transitions and missed imaging targets.

METHOD

- Fault-injected training: policies are trained inside a faulty environment.
- Fault information is added to the observation vector.
- Recurrent networks use temporal history to infer fault signatures.

Resilience must be designed into training — not added after it.

WHAT IT LEAVES OPEN

- it is a scheduler, not an attitude controller
- no continuous-control stability argument
- simulation-only evidence

RELEVANCE TO THIS THESIS

Directly motivates this thesis's train/test split: nominal vs fault-aware training, then unseen faults.

Nagano, Y. & Schaub, H. (2025). AAS 25-637, Enhancing Fault Resilience in RL-Based Satellite Autonomous Task Scheduling.

From fault-tolerant RL to trustworthy fault-tolerant RL

01 **Unseen faults**

Most results are evaluated on the fault set used for training.

→ Disjoint train / test fault distributions: severity, timing, combinations.

02 **Unverified actions**

A neural policy can leave the region where it was validated.

→ Physics-based governor: vet the action, otherwise hand over to LQR.

03 **Detection and control are separate**

Health estimation and the control law are designed apart.

→ Health estimate $\hat{h}$ inside the observation — no access to the true h.

04 **Deployment is assumed**

A policy that runs on a GPU is not a flight controller.

→ Latency budget, quantisation, and HIL on an MCU.

What I take away — and what I intend to prove

1. ADCS is mission-critical

The spacecraft must know and control its orientation, continuously.

2. Complexity follows the mission

Not every satellite needs the most advanced controller.

3. Faults are the design driver

On a nanosatellite, survival is a software property.

4. RL adds adaptability — and questions

Generalisation, safety, unseen faults, embedded cost.

THE GOAL

Not to replace a classical controller with a network — **but to find out whether learning can make attitude control more adaptive and fault-tolerant without giving up safety, reliability and deployability.**

SEMINAR I

Thank you for your attention

Questions and discussion

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Image: NASA/JPL-Caltech, NISAR in Earth orbit (artist's concept), image PIA26285.

APPENDIX

References

[1] Yang, S., Jin, L. & Rao, J. (2025). Reinforcement learning based attitude fault-tolerant control of spacecraft with unknown system model. Journal of the Franklin Institute, 362, 107741. doi:10.1016/j.jfranklin.2025.107741

[2] Lu, W., Zhuang, H., Geng, Z., Shen, Q., Mu, Z., Wu, S., Razoumny, V. Y. & Razoumny, Y. N. (2026). Safe reinforcement learning based agile satellite attitude control for non-cooperative target tracking with operation constraints and actuator faults. Aerospace Science and Technology, 168, 111238. doi:10.1016/j.ast.2025.111238

[3] Nagano, Y. & Schaub, H. (2025). Enhancing fault resilience in RL-based satellite autonomous task scheduling. AAS 25-637.

[4] NASA Small Spacecraft Systems Virtual Institute (2026). State-of-the-Art of Small Spacecraft Technology — Guidance, Navigation, and Control.

[5] NASA SSRI Knowledge Base. Attitude Determination and Control — Small Satellite Systems Engineering.

[6] Pouria Zarei (2026). M.Sc. thesis proposal: Fault-Tolerant Attitude Control of Nanosatellites Using Reinforcement Learning — From Simulation to Hardware-in-the-Loop Implementation. University of Tehran.

IMAGE CREDITS

All spacecraft imagery: NASA / JPL-Caltech, public domain (PIA26285, PIA22457).